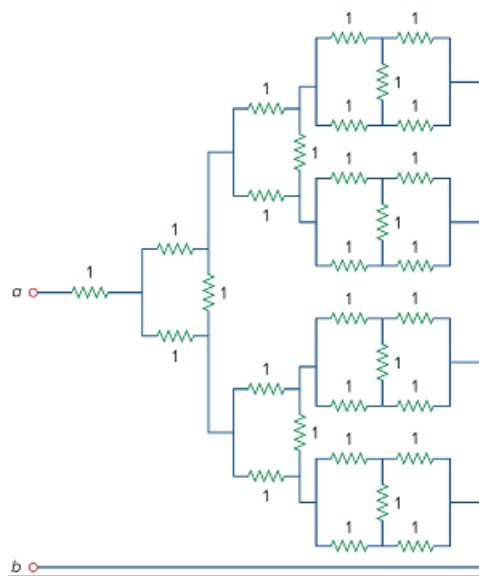


Problem

Repeat Prob. 2.75 for the eight-way divider shown in Fig. 2.136.

Figure 2.136:



Step-by-step solution

Step 1 of 13

Consider the following circuit diagram:

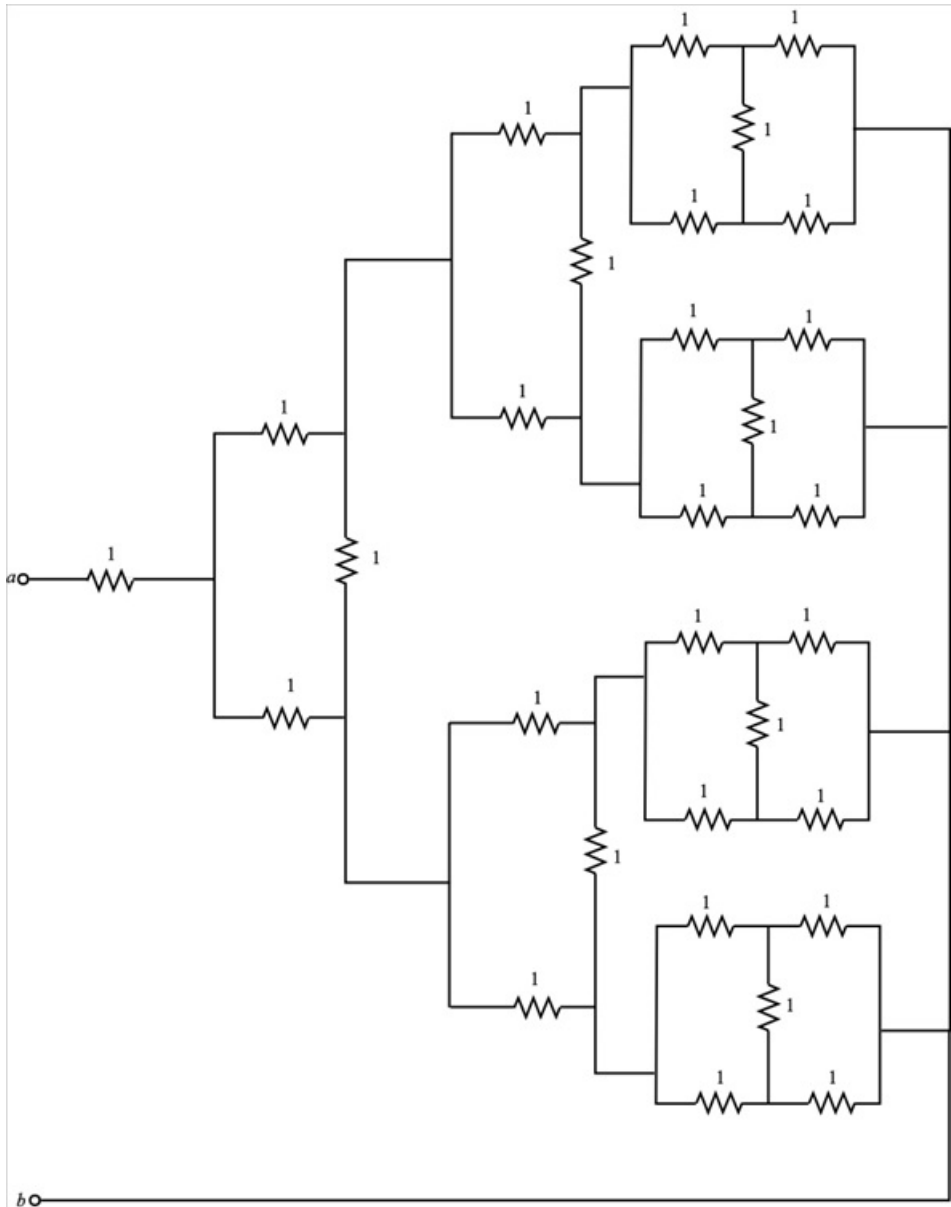


Figure 1

Consider the following sub diagram from Figure 1:

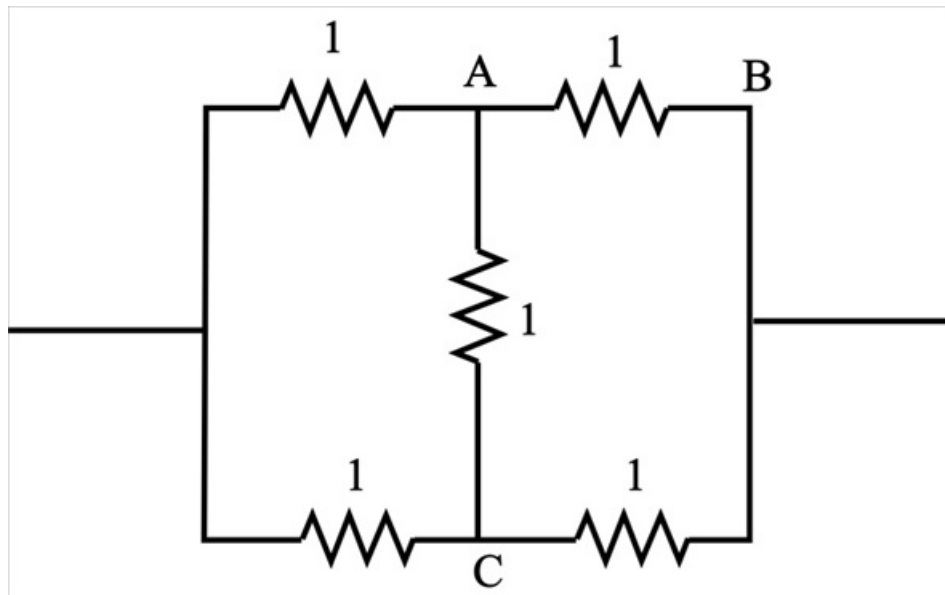


Figure 2

From Figure 2,

Apply Δ to Y transform between the points A, B and C.

$$\begin{aligned} R_1 &= \frac{R_B R_C}{R_A R_B + R_B R_C + R_C R_A} \\ &= \frac{(1)(1)}{(1)(1) + (1)(1) + (1)(1)} \\ &= \frac{1}{1+1+1} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} R_2 &= \frac{R_A R_C}{R_A R_B + R_B R_C + R_C R_A} \\ &= \frac{(1)(1)}{(1)(1) + (1)(1) + (1)(1)} \\ &= \frac{1}{1+1+1} \\ &= \frac{1}{3} \end{aligned}$$

$$\begin{aligned} R_3 &= \frac{R_A R_B}{R_A R_B + R_B R_C + R_C R_A} \\ &= \frac{(1)(1)}{(1)(1) + (1)(1) + (1)(1)} \\ &= \frac{1}{1+1+1} \\ &= \frac{1}{3} \end{aligned}$$

The modified diagram of Figure 2 is shown in Figure 3:

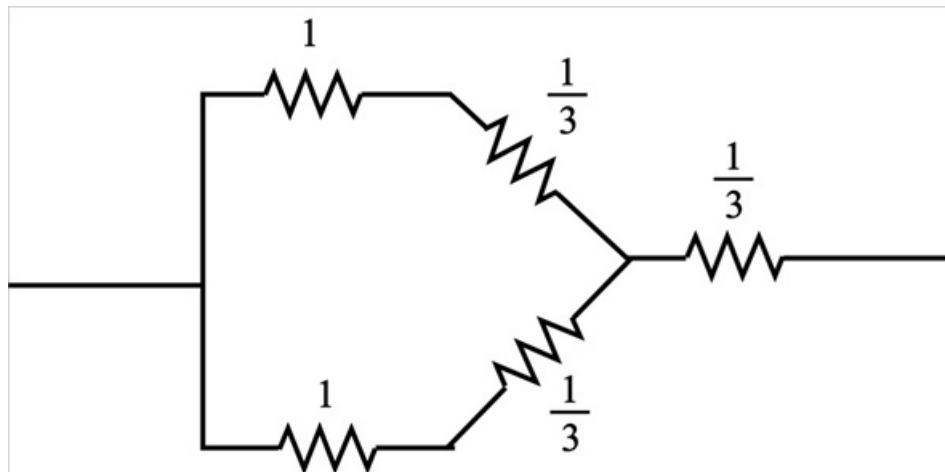


Figure 3

Step 4 of 13

From Figure 3, the resistors 1 and $\frac{1}{3}$ are in series. The equivalent resistance is calculated as follows.

$$\begin{aligned} \text{Equivalent resistance} &= 1 + \frac{1}{3} \\ &= \frac{4}{3} \end{aligned}$$

Step 5 of 13

The modified diagram of Figure 3 is shown in Figure 4:

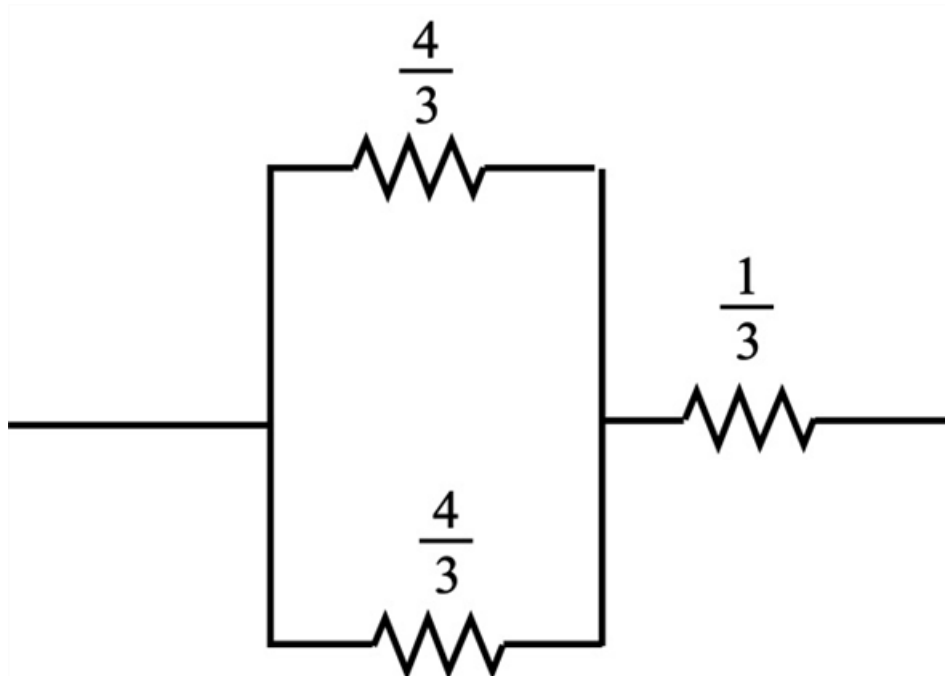


Figure 4

Step 6 of 13

From Figure 4, the resistances $\frac{4}{3}$ and $\frac{4}{3}$ are in parallel. The equivalent resistance is calculated as follows.

$$\begin{aligned} \left(\frac{4}{3}\right) \parallel \left(\frac{4}{3}\right) &= \frac{\left(\frac{4}{3}\right)\left(\frac{4}{3}\right)}{\left(\frac{4}{3}\right) + \left(\frac{4}{3}\right)} \\ &= \frac{\left(\frac{16}{9}\right)}{\left(\frac{8}{3}\right)} \\ &= \frac{2}{3} \end{aligned}$$

Step 7 of 13

The modified diagram of Figure 4 is shown in Figure 5:

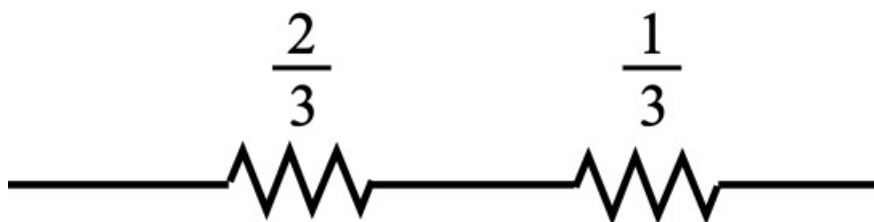


Figure 5

Step 8 of 13

From Figure 5, the resistors $\frac{2}{3}$ and $\frac{1}{3}$ are in series. Calculate the equivalent resistance is as follows.

$$\begin{aligned}\text{Equivalent resistance} &= \frac{2}{3} + \frac{1}{3} \\ &= \frac{3}{3} \\ &= 1\end{aligned}$$

The equivalent circuit of Figure 2 is shown in Figure 6:

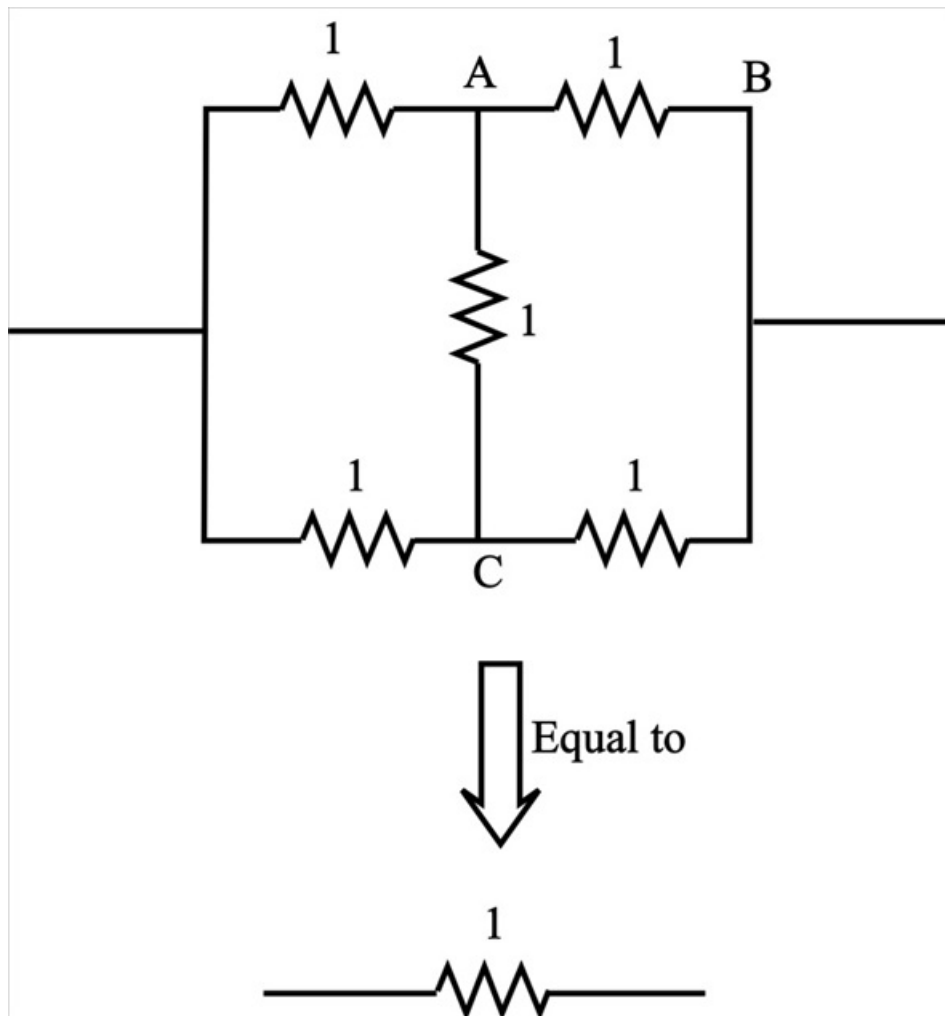


Figure 6

The modified diagram of Figure 1 is shown in Figure 7:

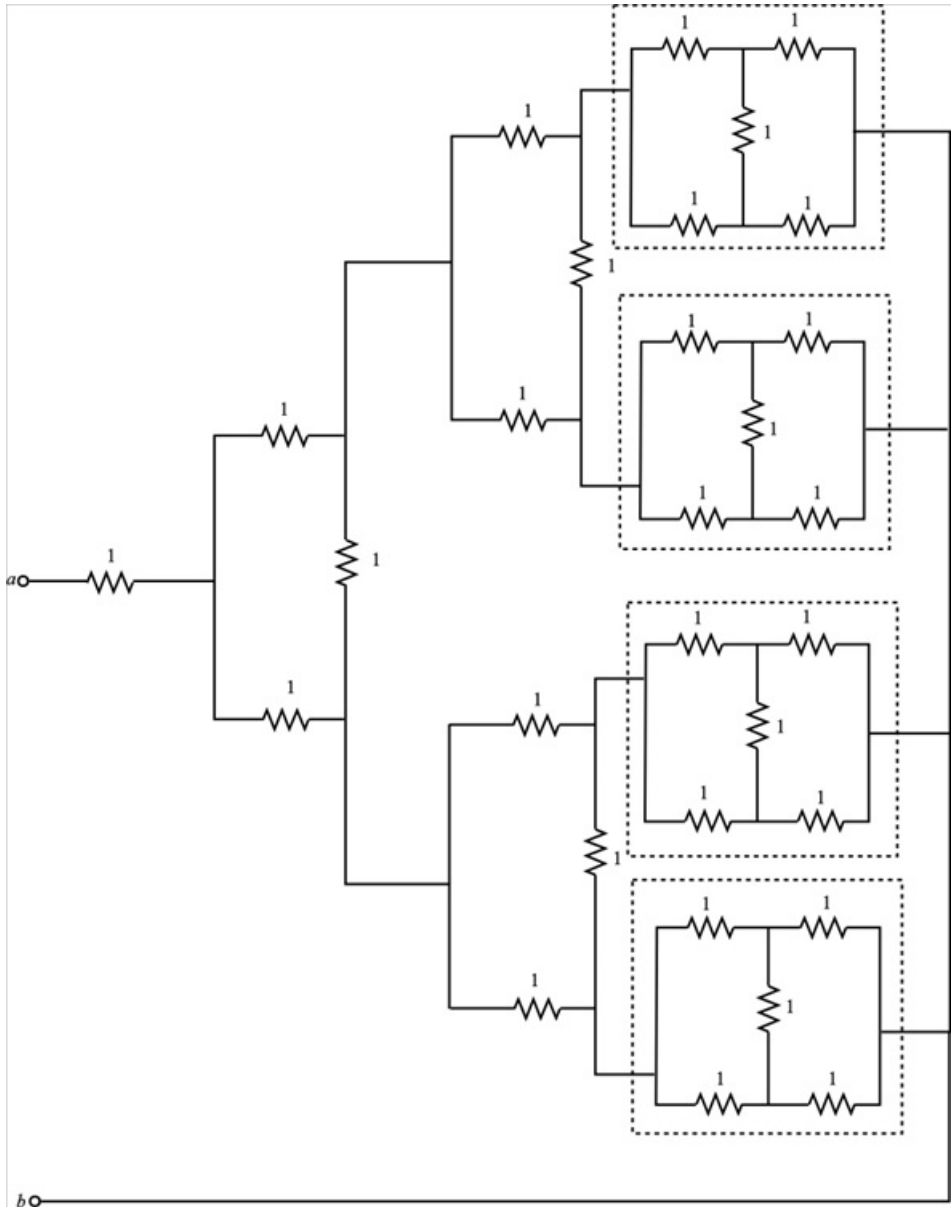


Figure 7

From Figure 7, the equivalent resistance 1 is replaced by the sub diagram as shown in Figure 6.

The modified diagram of Figure 7 is shown in Figure 8:

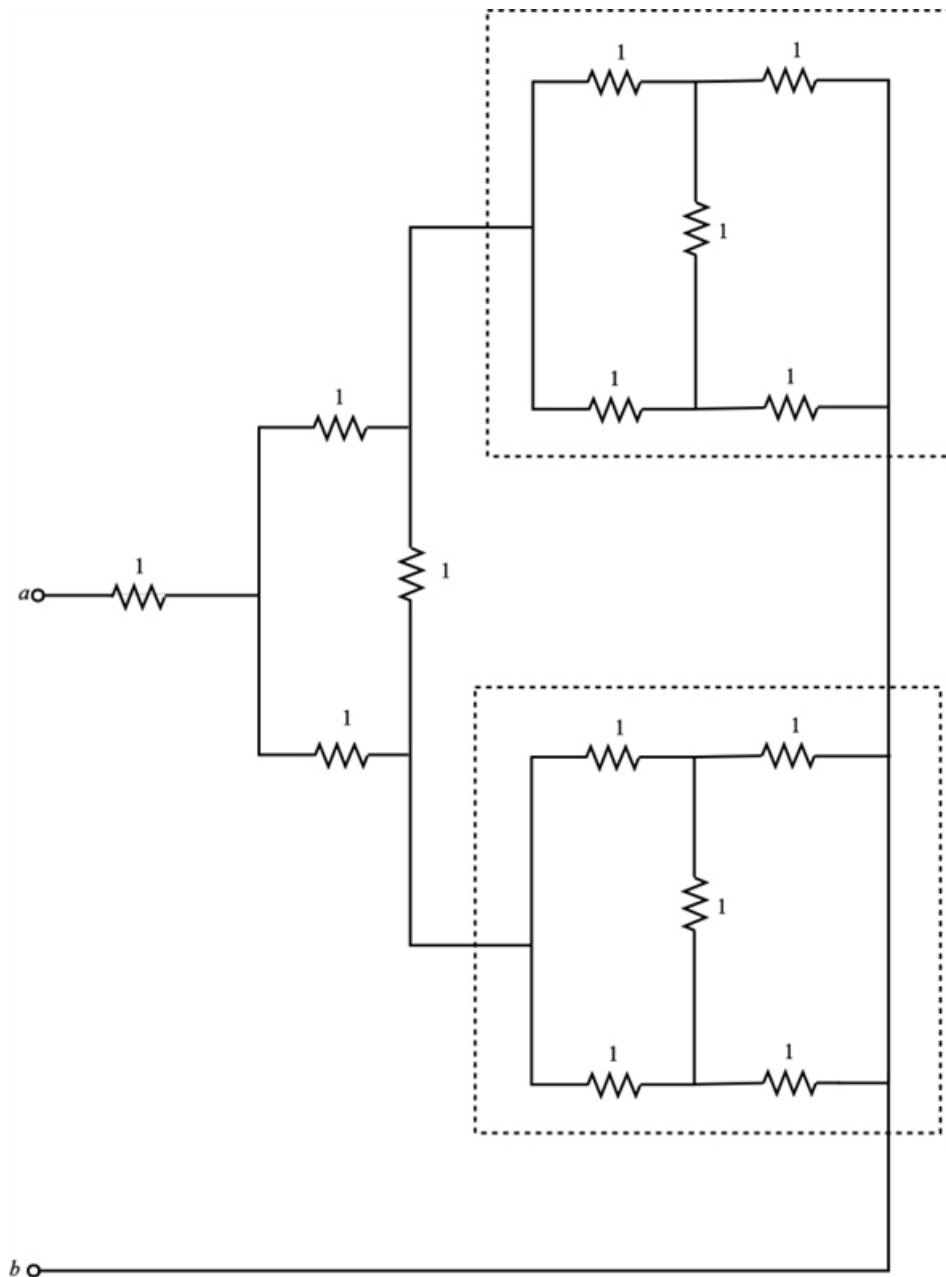


Figure 8

From Figure 8, the equivalent resistance 1 is replaced by the sub diagram shown in Figure 6.

The modified diagram of Figure 8 is shown in Figure 9:

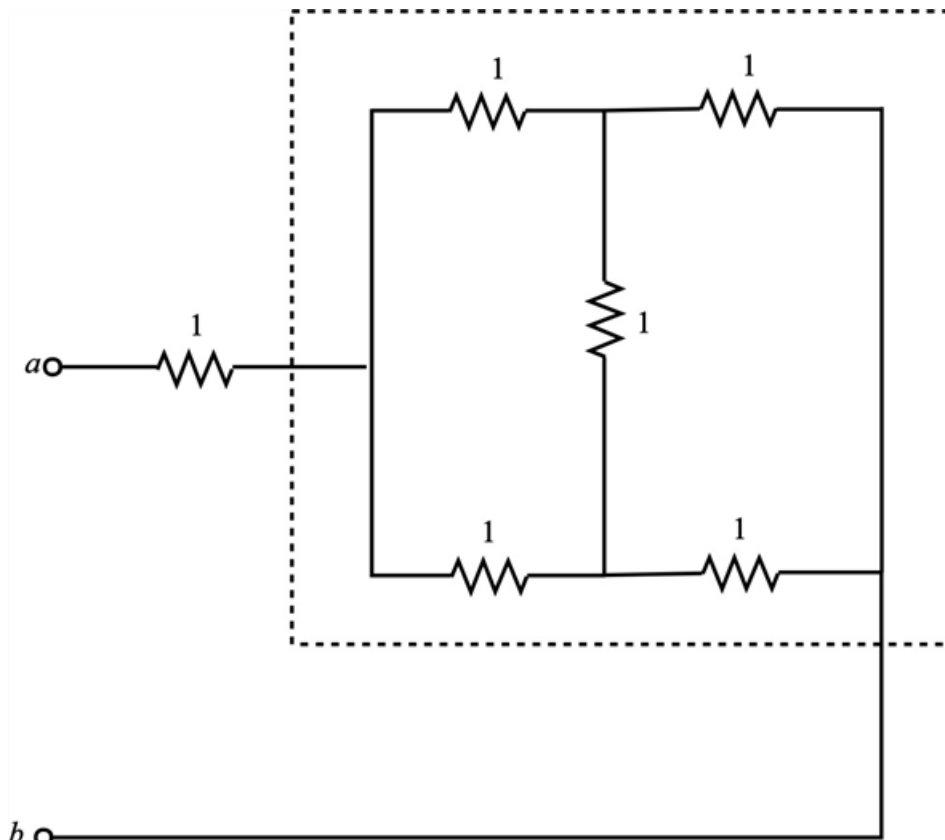


Figure 9

Step 12 of 13

From Figure 9, the equivalent resistance 1 is replaced by the sub diagram shown in Figure 6.

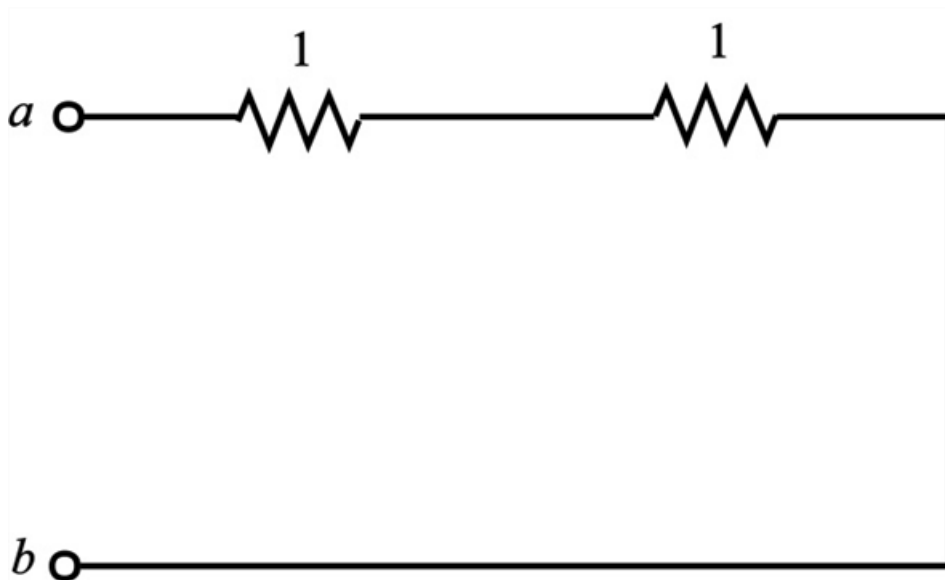


Figure 10

Step 13 of 13

From Figure 10, calculate the equivalent resistance between a and b is as follows.

$$\begin{aligned}\text{Equivalent resistance} &= 1 + 1 \\ &= 2\end{aligned}$$

Therefore, the equivalent resistance between a and b is 2Ω .